

Aditi Anand Jadhav

Full Stack Developer — CI/CD — Cloud Engineer (AWS/GCP)
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SUMMARY

Associate Software Development Engineer at Publicis Sapient with 1.6 years of experience in full-stack development using ReactJS, Spring Boot, Jenkins, and Docker. Experienced in CI/CD pipelines, containerization, and scalable web applications. Contributed to backend API development, frontend engineering, and cloud deployment. Led an AI-driven malware detection project, strengthening expertise in cybersecurity, machine learning, and data analytics. Thrive in dynamic environments with strong problem-solving, adaptability, and collaboration skills. Experienced in building scalable microservices, secure authentication systems, monitoring solutions, and production-ready deployments on AWS and GCP.

TECHNICAL SKILLS

Frontend: React.js, JavaScript, HTML5, CSS3, Lighthouse Audits

Backend: Spring Boot, REST APIs, FastAPI, GraphQL, Java, Python, SQL, Microservices Architecture, JWT Authentication, Role-Based Access Control (RBAC), PostgreSQL, Database Schema Design, ER Modeling

DevOps: Docker, Jenkins, GCP, Google Cloud Run, Artifact Registry, AWS, CI/CD, GitHub Actions, SonarQube, Prometheus, Grafana, Microservices Deployment

Testing: Unit Testing, BDD, TDD, Cucumber, Mockito, JUnit, JMeter

AI/ML: Agentic AI, Prompt Engineering, Large Language Models (LLMs), OpenAI APIs, CLIP Models, Computer Vision, Image Processing

EXPERIENCE

Publicis Sapient

On-Site

Associate Software Development Engineer 1

Photo Intelligence System — AI-Powered Image Analysis Platform

Present

- Designed an end-to-end AI-powered Photo Intelligence System for batch image analysis, semantic search, and automated photo organization.
- Built a full-stack system using React + TypeScript frontend and FastAPI backend with async background processing pipelines for scalable workflows.
- Developed computer vision pipelines for blur detection, quality scoring, duplicate detection, screenshot detection, face grouping, and event clustering using EXIF data.
- Integrated OpenCV, CLIP Transformers, imagehash, SSIM, and optional DeepFace for multimodal image understanding and clustering.
- Implemented semantic image search using CLIP embeddings and cosine similarity for natural language-based photo retrieval.
- Built near-duplicate detection using SHA-256 hashing, perceptual hashing (pHash), SSIM, and embedding similarity refinement.
- Used CLIP zero-shot learning for scene classification into categories like travel, food, indoor, outdoor, and screenshots.
- Implemented DBSCAN-based clustering for face grouping and EXIF-based event detection.
- Designed FastAPI REST APIs with async job processing, multipart uploads, semantic search, and PDF report generation.
- Built a React dashboard with analytics, charts, galleries, semantic search, and real-time job status tracking.
- Generated structured JSON and PDF reports using ReportLab for insights and analytics.
- Containerized using Docker and deployed on Google Cloud Run with autoscaling and CI/CD pipelines.

- Designed modular architecture with reusable ML services, job lifecycle management, and local persistence for scalable deployment.

Key Concepts: FastAPI, React.js, TypeScript, OpenCV, Pillow, CLIP, Hugging Face Transformers, Semantic Search, Image Embeddings, Perceptual Hashing (pHash), SSIM, DBSCAN Clustering, DeepFace, OCR, Async Processing, REST APIs, Swagger/OpenAPI, Docker, Google Cloud Run, GCP Artifact Registry, Computer Vision, AI/ML Pipelines, Semantic Image Search, Event Clustering, Face Recognition, PDF Report Generation, System Design, Full-Stack Development

Fraud Detection Service (Natwest Bank)

Feb 2026

- Designed and developed a full-stack Fraud Detection Service enabling users to report suspicious financial transactions and analysts to triage cases using automated risk scoring.
- Implemented JWT-based authentication and role-based access control (User / Analyst) using Spring Security to secure APIs and dashboards.
- Built RESTful APIs for case creation, status updates, audit logging, and risk scoring; documented endpoints using Swagger.
- Designed relational database schema for cases, notes, users, and audit logs using PostgreSQL with proper migration management.
- Developed rule-based Risk Scoring Engine to categorize cases into Low, Medium, and High risk buckets based on transaction patterns.
- Containerized backend and frontend services using Docker and deployed on Kubernetes (AWS EKS) with Application Load Balancer integration.
- Configured CI/CD pipeline using GitHub and AWS services to automate build, push, and deployment workflows.
- Integrated logging and monitoring to track authentication events, case updates, and system errors for production readiness.

Key Concepts: JWT Authentication, Role-Based Access Control (RBAC), Spring Security, RESTful APIs, Swagger, PostgreSQL, Database Design & Migrations, Risk Scoring Engine, Fraud Detection Logic, Docker, Kubernetes, AWS EKS, CI/CD Pipelines, GitHub Actions, Logging & Monitoring, System Design

Economic Crime Prevention Platform (Lloyds Bank)

April 2025 – July 2025

- Automated end-to-end testing using Playwright with BDD (Cucumber, Gherkin).
- Built reusable automation framework with XPath locators and CI/CD integration.
- Enhanced test coverage, early defect detection, and overall system reliability.

Key Concepts: Playwright, BDD, Cucumber, Gherkin, Test Automation, XPath, CI/CD, Test Framework Design

Banking Analytics Platform (Lloyds Bank)

Dec 2024 – Feb 2025

- Developed microservices-based analytics platform using Java, Spring Boot, and React with Figma-based UI/UX design and UML-based system design documentation.
- Implemented secure authentication, ETL pipelines, and transaction aggregation logic.
- Improved performance with API pagination and ensured quality via unit and UI testing.

Key Concepts: Microservices, Spring Boot, ReactJS, ETL Pipelines, Authentication, API Pagination, Data Processing, Cloud Deployment, GCP, System Design, Figma

PROJECTS

Android Botnet Detection Using AI

2023

- Built ML-based Android malware detection system achieving 92% accuracy.
- Reduced false positives by 30% using network traffic analysis techniques.
- Applied machine learning algorithms for botnet behavior detection.

Token System for Bank Operations

2021

- Developed queue-based token management system to streamline banking operations.
- Improved customer service efficiency by 25%.

EDUCATION

BTech – Information Technology - MKSSS's Cummins College of Engineering for Women, Pune	2020 – 2024
HSC – Indira Gandhi Junior College	2019 – 2020
SSC – Kusumtai High School, Nanded	2017 – 2018

CERTIFICATIONS

- Google Cloud Digital Leader – July 2025
- AWS Partner Accreditation (Technical) – Feb 2025
- Cloud Computing Program – IIT Bhubaneswar – May 2022
- Web Development Program – IIT Bombay – Feb 2022